SG No. 177 D_6^1 $P622$ [hexagonal]

* plus set: $+[0, 0, 0]$
* Wyckoff site: `1a`, site symmetry: `622`

Table 1: Wyckoff bond: `1a@1a`, bond symmetry: `622`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, 0]$	`[1,2,3,4,5,6,-7,-8,-9,-10,-11,-12]`

Table 2: Wyckoff bond: `3b@1a`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$[0, 0, 0]$	`[1,-4,-8,11]`
2	$[-2X, -X, 0]$	$[0, 0, 0]$	`[2,-5,-7,10]`
3	$[X, -X, 0]$	$[0, 0, 0]$	`[3,-6,-9,12]`

Table 3: Wyckoff bond: `3c@1a`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, 0, 0]$	`[1,-4,8,-11]`
2	$[0, X, 0]$	$[0, 0, 0]$	`[2,-5,7,-10]`
3	$[-X, -X, 0]$	$[0, 0, 0]$	`[3,-6,9,-12]`

Table 4: Wyckoff bond: `6d@1a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, 0]$	`[1,-4]`
2	$[-Y, X-Y, 0]$	$[0, 0, 0]$	`[2,-5]`
3	$[-X+Y, -X, 0]$	$[0, 0, 0]$	`[3,-6]`
4	$[Y, X, 0]$	$[0, 0, 0]$	`[7,-10]`
5	$[X-Y, -Y, 0]$	$[0, 0, 0]$	`[8,-11]`
6	$[-X, -X+Y, 0]$	$[0, 0, 0]$	`[9,-12]`

Table 5: Wyckoff bond: `6e@1a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$[0, 0, 0]$	`[1,-8]`
2	$[-2X, -X, Z]$	$[0, 0, 0]$	`[2,-7]`
3	$[X, -X, Z]$	$[0, 0, 0]$	`[3,-9]`
4	$[-X, -2X, Z]$	$[0, 0, 0]$	`[4,-11]`
5	$[2X, X, Z]$	$[0, 0, 0]$	`[5,-10]`
6	$[-X, X, Z]$	$[0, 0, 0]$	`[6,-12]`

Table 6: Wyckoff bond: `6f@1a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, 0]$	`[1,-11]`
2	$[0, X, Z]$	$[0, 0, 0]$	`[2,-10]`
3	$[-X, -X, Z]$	$[0, 0, 0]$	`[3,-12]`
4	$[-X, 0, Z]$	$[0, 0, 0]$	`[4,-8]`
5	$[0, -X, Z]$	$[0, 0, 0]$	`[5,-7]`
6	$[X, X, Z]$	$[0, 0, 0]$	`[6,-9]`

Table 7: Wyckoff bond: `12g@1a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, 0, 0]$	`[2]`
3	$[-X+Y, -X, Z]$	$[0, 0, 0]$	`[3]`
4	$[-X, -Y, Z]$	$[0, 0, 0]$	`[4]`
5	$[Y, -X+Y, Z]$	$[0, 0, 0]$	`[5]`
6	$[X-Y, X, Z]$	$[0, 0, 0]$	`[6]`
7	$[Y, X, -Z]$	$[0, 0, 0]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[0, 0, 0]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[0, 0, 0]$	`[9]`
10	$[-Y, -X, -Z]$	$[0, 0, 0]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[0, 0, 0]$	`[11]`
12	$[X, X-Y, -Z]$	$[0, 0, 0]$	`[12]`

* Wyckoff site: `1b`, site symmetry: `622`

Table 8: Wyckoff bond: `1a@1b`, bond symmetry: `622`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,-7,-8,-9,-10,-11,-12]`

Table 9: Wyckoff bond: `3b@1b`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-4,-8,11]`
2	$[-2X, -X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-5,-7,10]`
3	$[X, -X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-6,-9,12]`

Table 10: Wyckoff bond: `3c@1b`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-4,8,-11]`
2	$[0, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-5,7,-10]`
3	$[-X, -X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-6,9,-12]`

Table 11: Wyckoff bond: `6d@1b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, \frac{1}{2}]$	`[1,-4]`
2	$[-Y, X-Y, 0]$	$[0, 0, \frac{1}{2}]$	`[2,-5]`
3	$[-X+Y, -X, 0]$	$[0, 0, \frac{1}{2}]$	`[3,-6]`
4	$[Y, X, 0]$	$[0, 0, \frac{1}{2}]$	`[7,-10]`
5	$[X-Y, -Y, 0]$	$[0, 0, \frac{1}{2}]$	`[8,-11]`
6	$[-X, -X+Y, 0]$	$[0, 0, \frac{1}{2}]$	`[9,-12]`

Table 12: Wyckoff bond: `6e@1b`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$[0, 0, \frac{1}{2}]$	`[1,-8]`
2	$[-2X, -X, Z]$	$[0, 0, \frac{1}{2}]$	`[2,-7]`
3	$[X, -X, Z]$	$[0, 0, \frac{1}{2}]$	`[3,-9]`
4	$[-X, -2X, Z]$	$[0, 0, \frac{1}{2}]$	`[4,-11]`
5	$[2X, X, Z]$	$[0, 0, \frac{1}{2}]$	`[5,-10]`
6	$[-X, X, Z]$	$[0, 0, \frac{1}{2}]$	`[6,-12]`

Table 13: Wyckoff bond: `6f@1b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, \frac{1}{2}]$	`[1,-11]`
2	$[0, X, Z]$	$[0, 0, \frac{1}{2}]$	`[2,-10]`
3	$[-X, -X, Z]$	$[0, 0, \frac{1}{2}]$	`[3,-12]`
4	$[-X, 0, Z]$	$[0, 0, \frac{1}{2}]$	`[4,-8]`
5	$[0, -X, Z]$	$[0, 0, \frac{1}{2}]$	`[5,-7]`
6	$[X, X, Z]$	$[0, 0, \frac{1}{2}]$	`[6,-9]`

Table 14: Wyckoff bond: `12g@1b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, \frac{1}{2}]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, 0, \frac{1}{2}]$	`[2]`
3	$[-X+Y, -X, Z]$	$[0, 0, \frac{1}{2}]$	`[3]`
4	$[-X, -Y, Z]$	$[0, 0, \frac{1}{2}]$	`[4]`
5	$[Y, -X+Y, Z]$	$[0, 0, \frac{1}{2}]$	`[5]`
6	$[X-Y, X, Z]$	$[0, 0, \frac{1}{2}]$	`[6]`
7	$[Y, X, -Z]$	$[0, 0, \frac{1}{2}]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[9]`
10	$[-Y, -X, -Z]$	$[0, 0, \frac{1}{2}]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[11]`
12	$[X, X-Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[12]`

* Wyckoff site: `2c`, site symmetry: `3.2`

Table 15: Wyckoff bond: `2a@2c`, bond symmetry: `3.2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[\frac{1}{3},\ \frac{2}{3},\ 0]$	`[1,2,3,-10,-11,-12]`
2	$[0,\ 0,\ Z]$	$[\frac{2}{3},\ \frac{1}{3},\ 0]$	`[4,5,6,-7,-8,-9]`

Table 16: Wyckoff bond: `6b@2c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$[\frac{1}{3},\ \frac{2}{3},\ 0]$	`[1,-11]`
2	$[0,\ X,\ Z]$	$[\frac{1}{3},\ \frac{2}{3},\ 0]$	`[2,-10]`
3	$[-X,\ -X,\ Z]$	$[\frac{1}{3},\ \frac{2}{3},\ 0]$	`[3,-12]`
4	$[-X,\ 0,\ Z]$	$[\frac{2}{3},\ \frac{1}{3},\ 0]$	`[4,-8]`
5	$[0,\ -X,\ Z]$	$[\frac{2}{3},\ \frac{1}{3},\ 0]$	`[5,-7]`
6	$[X,\ X,\ Z]$	$[\frac{2}{3},\ \frac{1}{3},\ 0]$	`[6,-9]`

Table 17: Wyckoff bond: `6c@2c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 2X,\ 0]$	$[\frac{1}{3},\ \frac{2}{3},\ 0]$	`[1,11]`
2	$[-2X,\ -X,\ 0]$	$[\frac{1}{3},\ \frac{2}{3},\ 0]$	`[2,10]`
3	$[X,\ -X,\ 0]$	$[\frac{1}{3},\ \frac{2}{3},\ 0]$	`[3,12]`
4	$[-X,\ -2X,\ 0]$	$[\frac{2}{3},\ \frac{1}{3},\ 0]$	`[4,8]`
5	$[2X,\ X,\ 0]$	$[\frac{2}{3},\ \frac{1}{3},\ 0]$	`[5,7]`
6	$[-X,\ X,\ 0]$	$[\frac{2}{3},\ \frac{1}{3},\ 0]$	`[6,9]`

Table 18: Wyckoff bond: `12d@2c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[\frac{1}{3},\ \frac{2}{3},\ 0]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[\frac{1}{3},\ \frac{2}{3},\ 0]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[\frac{1}{3},\ \frac{2}{3},\ 0]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$[\frac{2}{3},\ \frac{1}{3},\ 0]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$[\frac{2}{3},\ \frac{1}{3},\ 0]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$[\frac{2}{3},\ \frac{1}{3},\ 0]$	`[6]`
7	$[Y,\ X,\ -Z]$	$[\frac{2}{3},\ \frac{1}{3},\ 0]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$[\frac{2}{3},\ \frac{1}{3},\ 0]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$[\frac{2}{3},\ \frac{1}{3},\ 0]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$[\frac{1}{3},\ \frac{2}{3},\ 0]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$[\frac{1}{3},\ \frac{2}{3},\ 0]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$[\frac{1}{3},\ \frac{2}{3},\ 0]$	`[12]`

* Wyckoff site: `2d`, site symmetry: `3.2`

Table 19: Wyckoff bond: `2a@2d`, bond symmetry: `3.2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[1,2,3,-10,-11,-12]`
2	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[4,5,6,-7,-8,-9]`

Table 20: Wyckoff bond: `6b@2d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[1,-11]`
2	$[0,\ X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[2,-10]`
3	$[-X,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[3,-12]`
4	$[-X,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[4,-8]`
5	$[0,\ -X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[5,-7]`
6	$[X,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[6,-9]`

Table 21: Wyckoff bond: `6c@2d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 2X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[1,11]`
2	$[-2X,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[2,10]`
3	$[X,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[3,12]`
4	$[-X,\ -2X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[4,8]`
5	$[2X,\ X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[5,7]`
6	$[-X,\ X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[6,9]`

Table 22: Wyckoff bond: `12d@2d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[12]`

* Wyckoff site: `2e`, site symmetry: `6..`

Table 23: Wyckoff bond: `2a@2e`, bond symmetry: `6..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ z]$	`[1,2,3,4,5,6]`
2	$[0,\ 0,\ -Z]$	$[0,\ 0,\ -z]$	`[7,8,9,10,11,12]`

Table 24: Wyckoff bond: `6b@2e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$[0,\ 0,\ z]$	`[1,-4]`
2	$[-Y,\ X-Y,\ 0]$	$[0,\ 0,\ z]$	`[2,-5]`
3	$[-X+Y,\ -X,\ 0]$	$[0,\ 0,\ z]$	`[3,-6]`
4	$[Y,\ X,\ 0]$	$[0,\ 0,\ -z]$	`[7,-10]`
5	$[X-Y,\ -Y,\ 0]$	$[0,\ 0,\ -z]$	`[8,-11]`
6	$[-X,\ -X+Y,\ 0]$	$[0,\ 0,\ -z]$	`[9,-12]`

Table 25: Wyckoff bond: `12c@2e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ z]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[0,\ 0,\ z]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[0,\ 0,\ z]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$[0,\ 0,\ z]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$[0,\ 0,\ z]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$[0,\ 0,\ z]$	`[6]`
7	$[Y,\ X,\ -Z]$	$[0,\ 0,\ -z]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$[0,\ 0,\ -z]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$[0,\ 0,\ -z]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$[0,\ 0,\ -z]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$[0,\ 0,\ -z]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$[0,\ 0,\ -z]$	`[12]`

* Wyckoff site: `3f`, site symmetry: `222`

Table 26: Wyckoff bond: `3a@3f`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X,\ 2X,\ 0]$	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,-4,-8,11]`
2	$[-2X,\ -X,\ 0]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,-5,-7,10]`
3	$[X,\ -X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,-6,-9,12]`

Table 27: Wyckoff bond: `3b@3f`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,-4,8,-11]`
2	$[0,\ X,\ 0]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,-5,7,-10]`

continued ...

Table 27

No.	vector	center	mapping
3	$[-X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-6,9,-12]`

Table 28: Wyckoff bond: `3c@3f`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,-8,-11]`
2	$[0, 0, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,5,-7,-10]`
3	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,6,-9,-12]`

Table 29: Wyckoff bond: `6d@3f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-4]`
2	$[-Y, X-Y, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-5]`
3	$[-X+Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-6]`
4	$[Y, X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[7,-10]`
5	$[X-Y, -Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[8,-11]`
6	$[-X, -X+Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-12]`

Table 30: Wyckoff bond: `6e@3f`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-8]`
2	$[-2X, -X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-7]`
3	$[X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-9]`
4	$[-X, -2X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-11]`
5	$[2X, X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[5,-10]`
6	$[-X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,-12]`

Table 31: Wyckoff bond: `6f@3f`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-11]`
2	$[0, X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-10]`
3	$[-X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-12]`
4	$[-X, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-8]`
5	$[0, -X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[5,-7]`
6	$[X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,-9]`

Table 32: Wyckoff bond: `12g@3f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{2}, 0, 0]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, \frac{1}{2}, 0]$	`[2]`
3	$[-X+Y, -X, Z]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[3]`
4	$[-X, -Y, Z]$	$[\frac{1}{2}, 0, 0]$	`[4]`
5	$[Y, -X+Y, Z]$	$[0, \frac{1}{2}, 0]$	`[5]`
6	$[X-Y, X, Z]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[6]`
7	$[Y, X, -Z]$	$[0, \frac{1}{2}, 0]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[\frac{1}{2}, 0, 0]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[9]`
10	$[-Y, -X, -Z]$	$[0, \frac{1}{2}, 0]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[\frac{1}{2}, 0, 0]$	`[11]`
12	$[X, X-Y, -Z]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[12]`

* Wyckoff site: `3g`, site symmetry: `222`

Table 33: Wyckoff bond: `3a@3g`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,-4,-8,11]`
2	$[-2X, -X, 0]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[2,-5,-7,10]`
3	$[X, -X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3,-6,-9,12]`

Table 34: Wyckoff bond: `3b@3g`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,-4,8,-11]`
2	$[0, X, 0]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[2,-5,7,-10]`
3	$[-X, -X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3,-6,9,-12]`

Table 35: Wyckoff bond: `3c@3g`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,4,-8,-11]`
2	$[0, 0, Z]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[2,5,-7,-10]`
3	$[0, 0, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3,6,-9,-12]`

Table 36: Wyckoff bond: `6d@3g`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,-4]`
2	$[-Y, X-Y, 0]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[2,-5]`
3	$[-X+Y, -X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3,-6]`
4	$[Y, X, 0]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[7,-10]`

continued ...

Table 36

No.	vector	center	mapping
5	$[X-Y,\,-Y,\,0]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[8,-11]`
6	$[-X,\,-X+Y,\,0]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[9,-12]`

Table 37: Wyckoff bond: `6e@3g`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\,2X,\,Z]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[1,-8]`
2	$[-2X,\,-X,\,Z]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$	`[2,-7]`
3	$[X,\,-X,\,Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[3,-9]`
4	$[-X,\,-2X,\,Z]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[4,-11]`
5	$[2X,\,X,\,Z]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$	`[5,-10]`
6	$[-X,\,X,\,Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[6,-12]`

Table 38: Wyckoff bond: `6f@3g`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\,0,\,Z]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[1,-11]`
2	$[0,\,X,\,Z]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$	`[2,-10]`
3	$[-X,\,-X,\,Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[3,-12]`
4	$[-X,\,0,\,Z]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[4,-8]`
5	$[0,\,-X,\,Z]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$	`[5,-7]`
6	$[X,\,X,\,Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[6,-9]`

Table 39: Wyckoff bond: `12g@3g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[1]`
2	$[-Y,\,X-Y,\,Z]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$	`[2]`
3	$[-X+Y,\,-X,\,Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[3]`
4	$[-X,\,-Y,\,Z]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[4]`
5	$[Y,\,-X+Y,\,Z]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$	`[5]`
6	$[X-Y,\,X,\,Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[6]`
7	$[Y,\,X,\,-Z]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$	`[7]`
8	$[X-Y,\,-Y,\,-Z]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[8]`
9	$[-X,\,-X+Y,\,-Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[9]`
10	$[-Y,\,-X,\,-Z]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$	`[10]`
11	$[-X+Y,\,Y,\,-Z]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[11]`
12	$[X,\,X-Y,\,-Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[12]`

* Wyckoff site: `4h`, site symmetry: `3..`

Table 40: Wyckoff bond: `4a@4h`, bond symmetry: `3..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3]`
2	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[4,5,6]`
3	$[0,\ 0,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7,8,9]`
4	$[0,\ 0,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[10,11,12]`

Table 41: Wyckoff bond: `12b@4h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[12]`

* Wyckoff site: `6i`, site symmetry: `2..`

Table 42: Wyckoff bond: `6a@6i`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,-4]`
2	$[-Y,\ X-Y,\ 0]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[2,-5]`
3	$[-X+Y,\ -X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[3,-6]`
4	$[Y,\ X,\ 0]$	$\left[0,\ \frac{1}{2},\ -z\right]$	`[7,-10]`
5	$[X-Y,\ -Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[8,-11]`
6	$[-X,\ -X+Y,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[9,-12]`

Table 43: Wyckoff bond: `6b@6i`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,4]`
2	$[0,\ 0,\ Z]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[2,5]`
3	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[3,6]`
4	$[0,\ 0,\ -Z]$	$\left[0,\ \frac{1}{2},\ -z\right]$	`[7,10]`
5	$[0,\ 0,\ -Z]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[8,11]`
6	$[0,\ 0,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[9,12]`

Table 44: Wyckoff bond: `12c@6i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3]`
4	$[-X, -Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[7]`
8	$[X-Y, -Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[8]`
9	$[-X, -X+Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[12]`

* Wyckoff site: `6j`, site symmetry: `.2.`

Table 45: Wyckoff bond: `6a@6j`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$[x, 0, 0]$	`[1,-8]`
2	$[-2X, -X, Z]$	$[0, x, 0]$	`[2,-7]`
3	$[X, -X, Z]$	$[-x, -x, 0]$	`[3,-9]`
4	$[-X, -2X, Z]$	$[-x, 0, 0]$	`[4,-11]`
5	$[2X, X, Z]$	$[0, -x, 0]$	`[5,-10]`
6	$[-X, X, Z]$	$[x, x, 0]$	`[6,-12]`

Table 46: Wyckoff bond: `6b@6j`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,8]`
2	$[0, X, 0]$	$[0, x, 0]$	`[2,7]`
3	$[-X, -X, 0]$	$[-x, -x, 0]$	`[3,9]`
4	$[-X, 0, 0]$	$[-x, 0, 0]$	`[4,11]`
5	$[0, -X, 0]$	$[0, -x, 0]$	`[5,10]`
6	$[X, X, 0]$	$[x, x, 0]$	`[6,12]`

Table 47: Wyckoff bond: `12c@6j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, x, 0]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-x, -x, 0]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, 0, 0]$	`[4]`
5	$[Y, -X+Y, Z]$	$[0, -x, 0]$	`[5]`
6	$[X-Y, X, Z]$	$[x, x, 0]$	`[6]`

continued ...

Table 47

No.	vector	center	mapping
7	$[Y, X, -Z]$	$[0, x, 0]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[x, 0, 0]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[-x, -x, 0]$	`[9]`
10	$[-Y, -X, -Z]$	$[0, -x, 0]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[-x, 0, 0]$	`[11]`
12	$[X, X-Y, -Z]$	$[x, x, 0]$	`[12]`

* Wyckoff site: `6k`, site symmetry: `.2.`

Table 48: Wyckoff bond: `6a@6k`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,-8]`
2	$[-2X, -X, Z]$	$\left[0, x, \frac{1}{2}\right]$	`[2,-7]`
3	$[X, -X, Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[3,-9]`
4	$[-X, -2X, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[4,-11]`
5	$[2X, X, Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[5,-10]`
6	$[-X, X, Z]$	$\left[x, x, \frac{1}{2}\right]$	`[6,-12]`

Table 49: Wyckoff bond: `6b@6k`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,8]`
2	$[0, X, 0]$	$\left[0, x, \frac{1}{2}\right]$	`[2,7]`
3	$[-X, -X, 0]$	$\left[-x, -x, \frac{1}{2}\right]$	`[3,9]`
4	$[-X, 0, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	`[4,11]`
5	$[0, -X, 0]$	$\left[0, -x, \frac{1}{2}\right]$	`[5,10]`
6	$[X, X, 0]$	$\left[x, x, \frac{1}{2}\right]$	`[6,12]`

Table 50: Wyckoff bond: `12c@6k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[0, x, \frac{1}{2}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[3]`
4	$[-X, -Y, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[x, x, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, x, \frac{1}{2}\right]$	`[7]`
8	$[X-Y, -Y, -Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[8]`
9	$[-X, -X+Y, -Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[x, x, \frac{1}{2}\right]$	`[12]`

* Wyckoff site: `6l`, site symmetry: `..2`

Table 51: Wyckoff bond: `6a@6l`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$[x, -x, 0]$	`[1,-10]`
2	$[-X, 0, Z]$	$[x, 2x, 0]$	`[2,-12]`
3	$[0, -X, Z]$	$[-2x, -x, 0]$	`[3,-11]`
4	$[-X, -X, Z]$	$[-x, x, 0]$	`[4,-7]`
5	$[X, 0, Z]$	$[-x, -2x, 0]$	`[5,-9]`
6	$[0, X, Z]$	$[2x, x, 0]$	`[6,-8]`

Table 52: Wyckoff bond: `6b@6l`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, -x, 0]$	`[1,10]`
2	$[X, 2X, 0]$	$[x, 2x, 0]$	`[2,12]`
3	$[-2X, -X, 0]$	$[-2x, -x, 0]$	`[3,11]`
4	$[-X, X, 0]$	$[-x, x, 0]$	`[4,7]`
5	$[-X, -2X, 0]$	$[-x, -2x, 0]$	`[5,9]`
6	$[2X, X, 0]$	$[2x, x, 0]$	`[6,8]`

Table 53: Wyckoff bond: `12c@6l`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, -x, 0]$	`[1]`
2	$[-Y, X-Y, Z]$	$[x, 2x, 0]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-2x, -x, 0]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, x, 0]$	`[4]`
5	$[Y, -X+Y, Z]$	$[-x, -2x, 0]$	`[5]`
6	$[X-Y, X, Z]$	$[2x, x, 0]$	`[6]`
7	$[Y, X, -Z]$	$[-x, x, 0]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[2x, x, 0]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[-x, -2x, 0]$	`[9]`
10	$[-Y, -X, -Z]$	$[x, -x, 0]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[-2x, -x, 0]$	`[11]`
12	$[X, X-Y, -Z]$	$[x, 2x, 0]$	`[12]`

* Wyckoff site: `6m`, site symmetry: `..2`

Table 54: Wyckoff bond: `6a@6m`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[x, -x, \frac{1}{2}\right]$	`[1,-10]`
2	$[-X, 0, Z]$	$\left[x, 2x, \frac{1}{2}\right]$	`[2,-12]`
3	$[0, -X, Z]$	$\left[-2x, -x, \frac{1}{2}\right]$	`[3,-11]`
4	$[-X, -X, Z]$	$\left[-x, x, \frac{1}{2}\right]$	`[4,-7]`
5	$[X, 0, Z]$	$\left[-x, -2x, \frac{1}{2}\right]$	`[5,-9]`

continued ...

Table 54

No.	vector	center	mapping
6	$[0,\, X,\, Z]$	$\left[2x,\, x,\, \frac{1}{2}\right]$	[6,-8]

Table 55: Wyckoff bond: 6b@6m, bond symmetry: ..2

No.	vector	center	mapping
1	$[X,\, -X,\, 0]$	$\left[x,\, -x,\, \frac{1}{2}\right]$	[1,10]
2	$[X,\, 2X,\, 0]$	$\left[x,\, 2x,\, \frac{1}{2}\right]$	[2,12]
3	$[-2X,\, -X,\, 0]$	$\left[-2x,\, -x,\, \frac{1}{2}\right]$	[3,11]
4	$[-X,\, X,\, 0]$	$\left[-x,\, x,\, \frac{1}{2}\right]$	[4,7]
5	$[-X,\, -2X,\, 0]$	$\left[-x,\, -2x,\, \frac{1}{2}\right]$	[5,9]
6	$[2X,\, X,\, 0]$	$\left[2x,\, x,\, \frac{1}{2}\right]$	[6,8]

Table 56: Wyckoff bond: 12c@6m, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$\left[x,\, -x,\, \frac{1}{2}\right]$	[1]
2	$[-Y,\, X-Y,\, Z]$	$\left[x,\, 2x,\, \frac{1}{2}\right]$	[2]
3	$[-X+Y,\, -X,\, Z]$	$\left[-2x,\, -x,\, \frac{1}{2}\right]$	[3]
4	$[-X,\, -Y,\, Z]$	$\left[-x,\, x,\, \frac{1}{2}\right]$	[4]
5	$[Y,\, -X+Y,\, Z]$	$\left[-x,\, -2x,\, \frac{1}{2}\right]$	[5]
6	$[X-Y,\, X,\, Z]$	$\left[2x,\, x,\, \frac{1}{2}\right]$	[6]
7	$[Y,\, X,\, -Z]$	$\left[-x,\, x,\, \frac{1}{2}\right]$	[7]
8	$[X-Y,\, -Y,\, -Z]$	$\left[2x,\, x,\, \frac{1}{2}\right]$	[8]
9	$[-X,\, -X+Y,\, -Z]$	$\left[-x,\, -2x,\, \frac{1}{2}\right]$	[9]
10	$[-Y,\, -X,\, -Z]$	$\left[x,\, -x,\, \frac{1}{2}\right]$	[10]
11	$[-X+Y,\, Y,\, -Z]$	$\left[-2x,\, -x,\, \frac{1}{2}\right]$	[11]
12	$[X,\, X-Y,\, -Z]$	$\left[x,\, 2x,\, \frac{1}{2}\right]$	[12]

* Wyckoff site: 12n, site symmetry: 1

Table 57: Wyckoff bond: 12a@12n, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$[x,\, y,\, z]$	[1]
2	$[-Y,\, X-Y,\, Z]$	$[-y,\, x-y,\, z]$	[2]
3	$[-X+Y,\, -X,\, Z]$	$[-x+y,\, -x,\, z]$	[3]
4	$[-X,\, -Y,\, Z]$	$[-x,\, -y,\, z]$	[4]
5	$[Y,\, -X+Y,\, Z]$	$[y,\, -x+y,\, z]$	[5]
6	$[X-Y,\, X,\, Z]$	$[x-y,\, x,\, z]$	[6]
7	$[Y,\, X,\, -Z]$	$[y,\, x,\, -z]$	[7]
8	$[X-Y,\, -Y,\, -Z]$	$[x-y,\, -y,\, -z]$	[8]
9	$[-X,\, -X+Y,\, -Z]$	$[-x,\, -x+y,\, -z]$	[9]
10	$[-Y,\, -X,\, -Z]$	$[-y,\, -x,\, -z]$	[10]
11	$[-X+Y,\, Y,\, -Z]$	$[-x+y,\, y,\, -z]$	[11]

continued ...

Table 57

No.	vector	center	mapping
12	$[X,\ X-Y,\ -Z]$	$[x,\ x-y,\ -z]$	[12]